

Deblurring and Denoising of Retinal Images Using Computational Imaging Methods

Jacopo Bonifazi - 0001217841
Gaetano Muscarello - 0001218184
Shivam Kumar - 0001242586

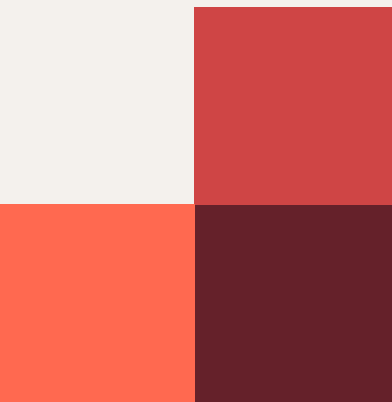
[Link repository](#)



Project description

The task must be formulated as an inverse problem where a degraded observation is mapped back to a high-quality reconstruction.

- **Target Task:** Deblur and Denoise
- **Methodological Requirements:**
 - Variational method: Regularization with Fista Wavelets
 - End-to-end method: UNet
 - Generative method: Diffusion with DiffPIR
 - Hybrid method: Weighted Total Variation
- **Deliverables:** PSNR and SSIM metrics and comparison plot



Methodology

Variational & Hybrid methods

Deblur + denoise as a model-based inverse problem

$$\min \frac{1}{2} \|A x - y\|^2 + \lambda R(x)$$

- **The degraded image is the blur applied to the clean image, plus some noise: $y = A x + e$**
- A = motion blur operator, ill-conditioned $\rightarrow$ naive inversion amplifies the noise
- Solved in model-based form: minimise data fidelity + regularisation
 - $\frac{1}{2}\|Ax - y\|^2 \rightarrow$ data-fidelity term (ℓ^2 , because the noise is Gaussian)
 - $R(x) \rightarrow$ regulariser, i.e. the prior knowledge we put on the image
 - $\lambda \rightarrow$ trade-off: too small leaves noise, too large over-smooths
- Both methods share this exact formulation - only the prior $R(x)$ changes

Fista + Wavelet

$$R(x) = \|W x\|_1 \quad (W = \text{wavelet transform, Daubechies db4, 3 levels})$$

- **Prior: natural images are sparse in the wavelet domain (few large coefficients)**
- Solved with FISTA algorithm - an accelerated proximal-gradient method
 - Each iteration = gradient step on the data term + proximal step
 - Proximal step = soft-thresholding of the wavelet coefficients (threshold λ/L)
 - Momentum (Nesterov acceleration) $\rightarrow$ convergence rate $O(1/k^2)$
- Step size $L = 1$, since $\|A^T A\| \approx 1$
- Convex problem $\rightarrow$ one single best solution; ~ 300 iterations



Adaptive Weighted TV

$$R(x) = \| w \odot |\nabla x| \|_1 \quad (\text{weighted total variation})$$

- **Prior: the image is piecewise-smooth in flat regions with sharp edges (ideal for vessels)**
- Spatially-varying weights w : small on edges, large on flat areas
 - Small weight on an edge $\rightarrow$ penalty relaxed $\rightarrow$ the edge is preserved
- **Weights computed from an intermediate image rough, then kept fixed**
 - We report TV- $W\ell_1$, where the rough image comes from a quick TV reconstruction.
 - Not iterative reweighting(computed once, then fixed)
- Solver: FISTA on smoothed TV (total variation has a sharp corner at zero)

End-to-end method

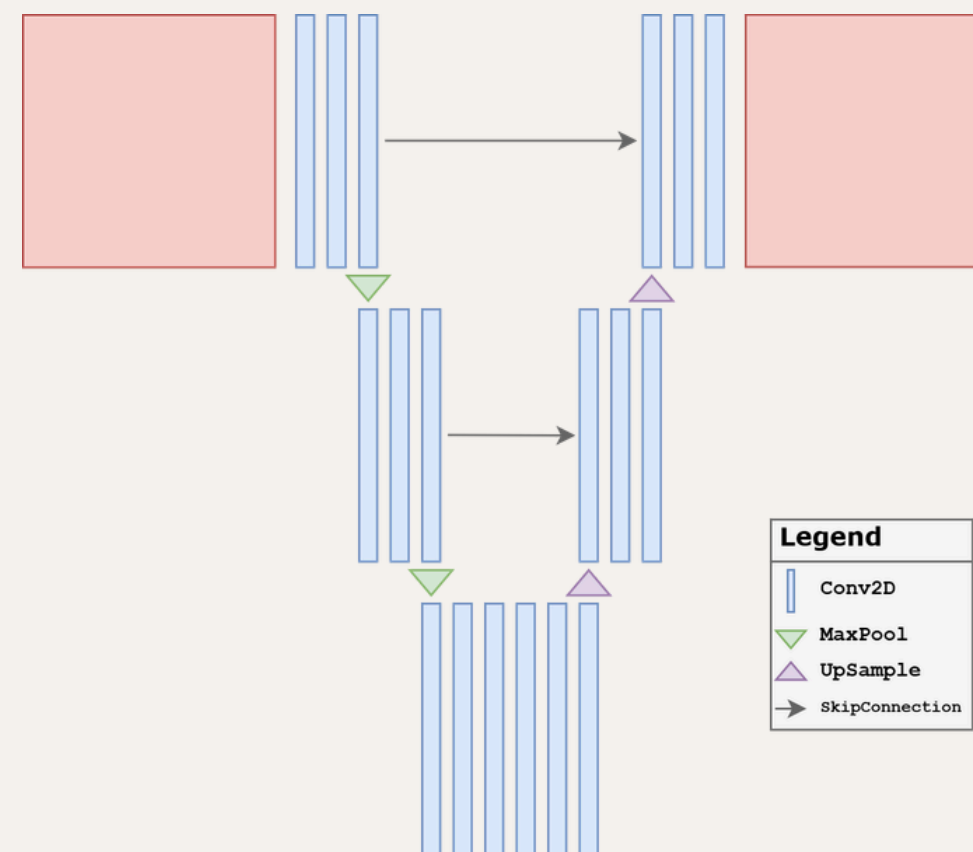
- End-to-end method works by collecting a training set of examples, comprising couples of associate images, and learn a parameterized inverse map.
- **Goal:** minimizing loss function that measures the discrepancy between the network output and the ground-truth target.
- **Limit:** even when they produce visually impressive reconstructions, they do not explicitly enforce data consistency with the measurement model.

$$f_{\Theta}(y^{\delta}) \mapsto x^{*}$$



End-to-end method - UNet

- **Encoder:** progressively reduces the spatial resolution through downsampling while increasing the number of feature channels.
- **Decoder:** progressively restores the spatial resolution until the output has the same size as the input.
- **Skip connections:** transfer feature maps from the encoder directly to the decoder at the same resolution level.



Generative Method - DiffPIR

- **Plug and Play approach:** a diffusion model acts as a *learned prior*, trained on clean retinal images to capture how a real retina looks like.
- Reconstruction alternates two steps at each interaction
 - **Denoiser step:** the diffusion prior pushes the estimate toward a plausible image
 - **Data-consistency step:** a correction toward the measurement, using the forward operator K and its adjoint K^T
- The parameter τ balances the two steps: high τ trusts the data, low τ trusts the prior.
- The prior is only trained **once** and reused for every noise level **without retraining**



Implementation

Algorithm steps

FISTA + Wavelet

- **Start** - first guess from the data. We push the measured image back through the operator
 $-x_0 = A^T y$
- **Gradient step** - move toward the data. We measure the mismatch $Az - y$ and step against its gradient.
 $-z = (1/L) \cdot A^T(Az - y)$
- **Clean the coefficients** - go to wavelet domain, shrink small coefficients to zero (soft-threshold). This removes noise.
- **Accelerate** (momentum) - combine the last two estimates to converge faster (Nesterov).
 $-z = x_k + (t_k - 1)/t_{k+1} \cdot (x_k - x_{k-1})$
- **Repeat** - about 300 iterations.

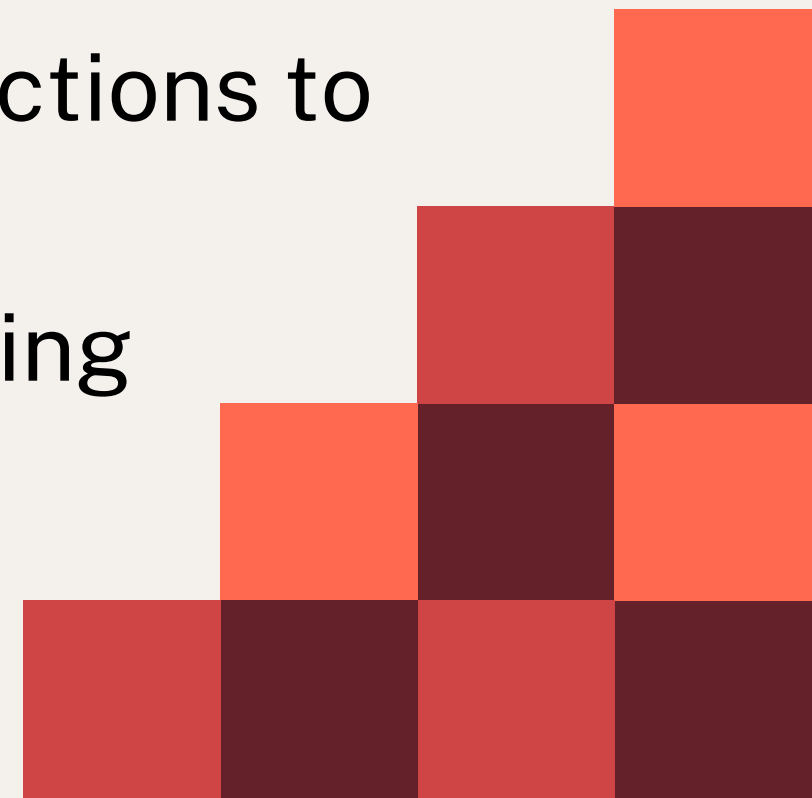
Adaptive Weighted TV

- **Rough image $\tilde{x}$** - a quick, early-stopped reconstruction. Not the final image; used only to locate edges. From a short TV (our version) or from a U-Net (Net-WL1).
 $-\tilde{x}$ = short reconstruction of y
- **Weights** (once, then fixed) - from the gradient of $\tilde{x}$, compute the weight map. Small on edges, large on flat areas.
 $w = (\eta / \sqrt{(\eta^2 + |\nabla \tilde{x}|^2)})^{(1-p)}$
- **Solve** - run FISTA again, now on the weighted, smoothed total variation. The weights say where to protect and where to smooth.
 $-\min \frac{1}{2} \|Ax - y\|^2 + \lambda \cdot \sum w \cdot \sqrt{(|\nabla x|^2 + \epsilon^2)}$

Shared setup: operator A = IPPy library, motion blur · images grayscale 256×256
 · λ chosen on the validation set · solver FISTA on smoothed TV.

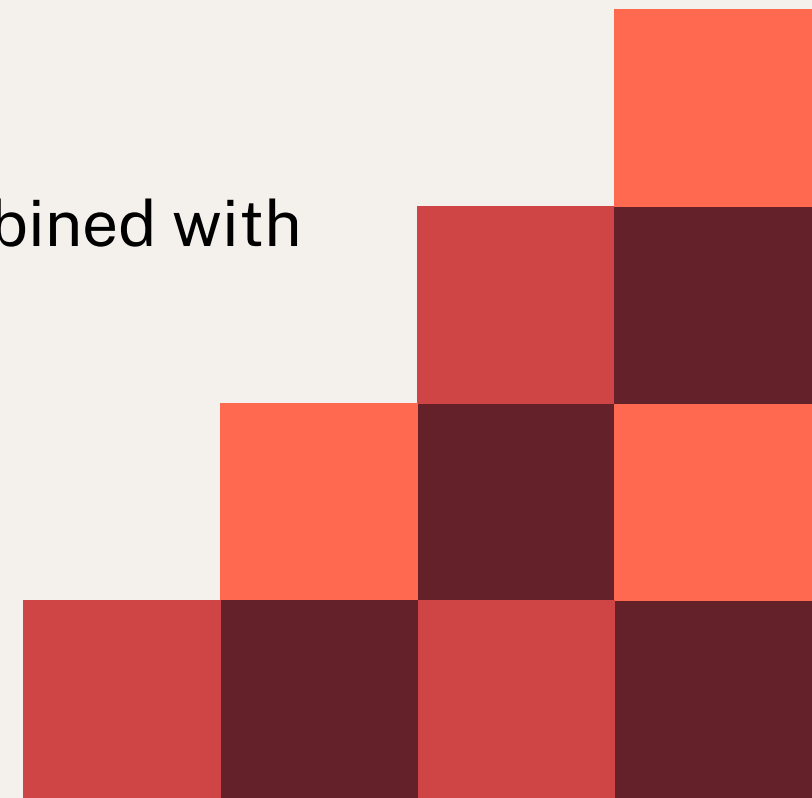
UNet

- The problem is treated as supervised image reconstruction: degraded retinal images are mapped to clean ground truth images.
- The dataset is loaded from precomputed .pt files containing paired degraded and clean tensors.
- Images are grayscale, resized to 256 x 256, normalized in [0, 1], and evaluated for four noise levels.
- A separate UNet is trained for each noise level using MSELoss.
- The UNet uses an encoder-decoder structure with skip connections to recover both global structure and fine details.
- The best validation checkpoint is evaluated on the test set using PSNR, SSIM, and visual comparisons.



DiffPIR

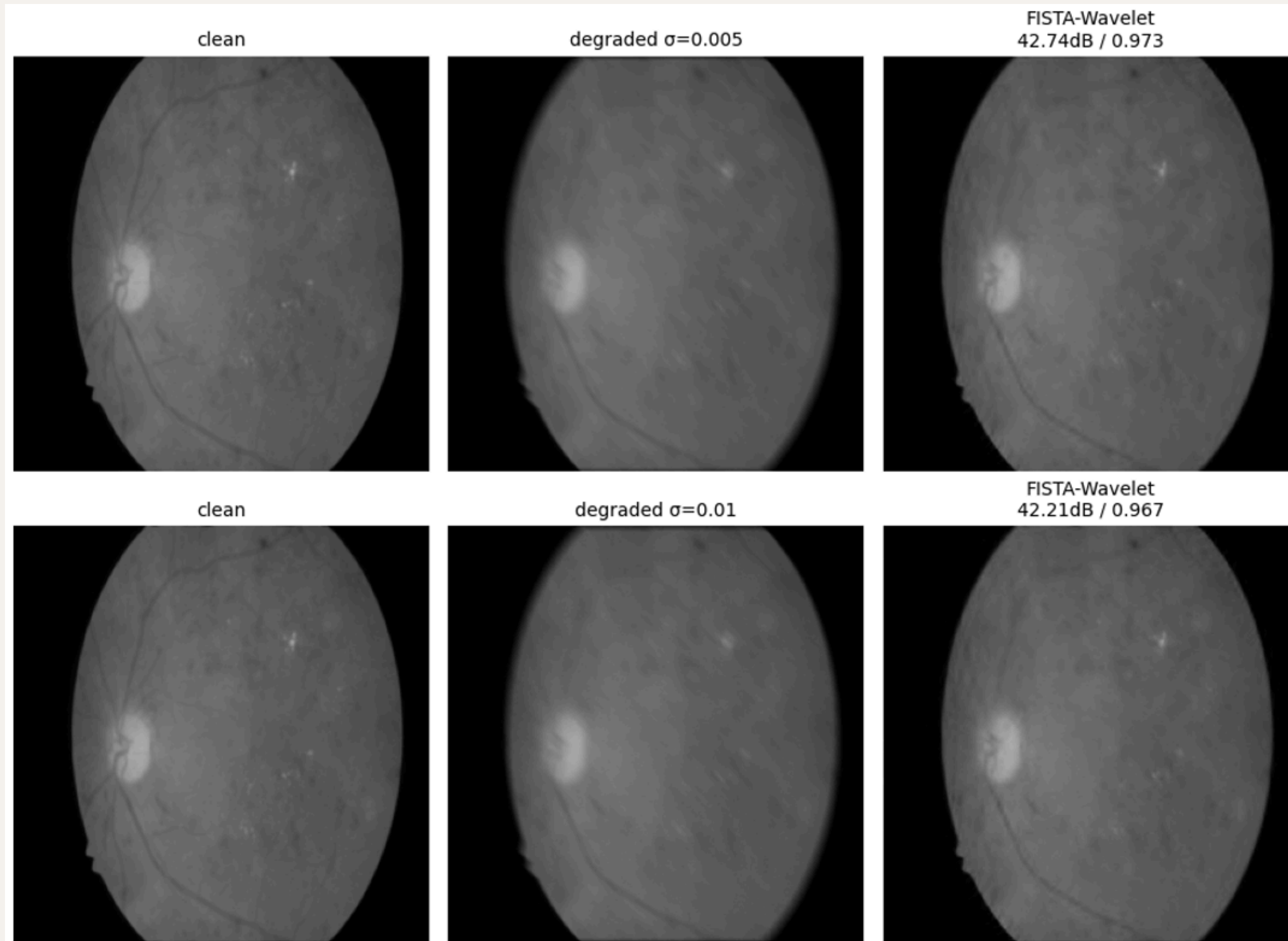
- The diffusion denoiser is trained on clean retinal images (256×256 , *grayscale*) to learn the image prior, during this phase the blur operator and noise are **not used**, they only enter at reconstruction time
- **Architecture adapted to 256×256 :** one extra downsampling level and self-attention only at low resolution (32×32), in order to keep memory usage feasible on the available GPU.
- Cosine noise schedule with 400 diffusion steps: reconstruction performed with 40 deterministic DDIM steps.
- At reconstruction, the motion blur operator (angle 45° , kernel size 9) and the noise are combined with the prior through the DiffPIR data-consistency update.



Results

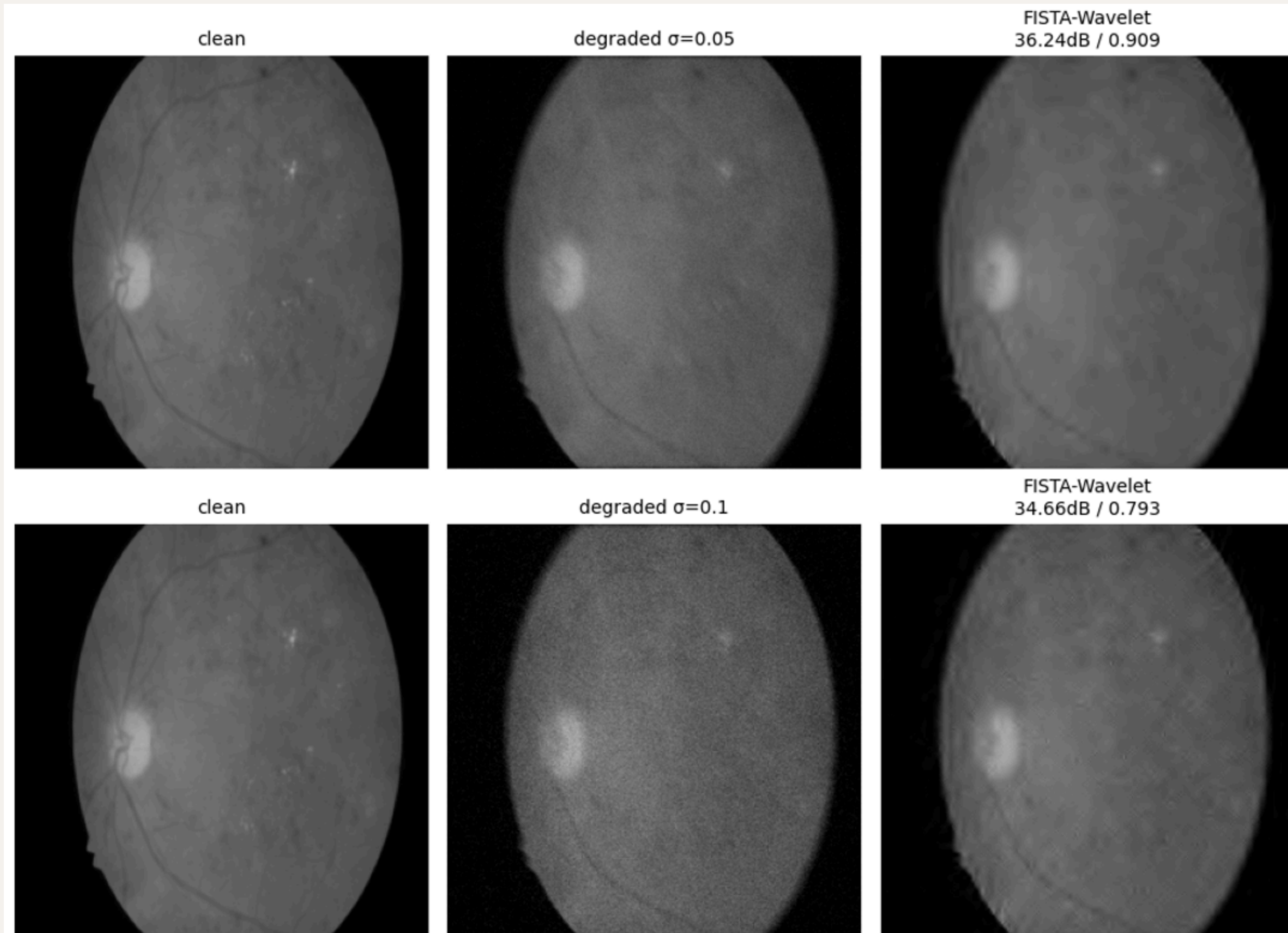
FISTA-Wavelet- Visual results

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FISTA-Wavelet- Visual results

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Noise	λ	PSNR	SSIM
0.005	0.001	41.44 $\pm$ 2.46	0.968 $\pm$ 0.013
0.01	0.001	40.95 $\pm$ 2.57	0.961 $\pm$ 0.017
0.05	0.02	34.53 $\pm$ 2.53	0.891 $\pm$ 0.039
0.1	0.02	33.49 $\pm$ 2.80	0.800 $\pm$ 0.091

Expected Monotonic Degradation

- PSNR drops from ~41 dB ($\lambda = 0.005$) to ~33 dB ($\lambda = 0.1$).
- SSIM decreases from 0.97 to 0.80.
- Low Noise Performance: Near-perfect image recovery (SSIM $\approx$ 0.97) validates wavelet sparsity as an effective prior for this dataset.

Impact of Higher Noise

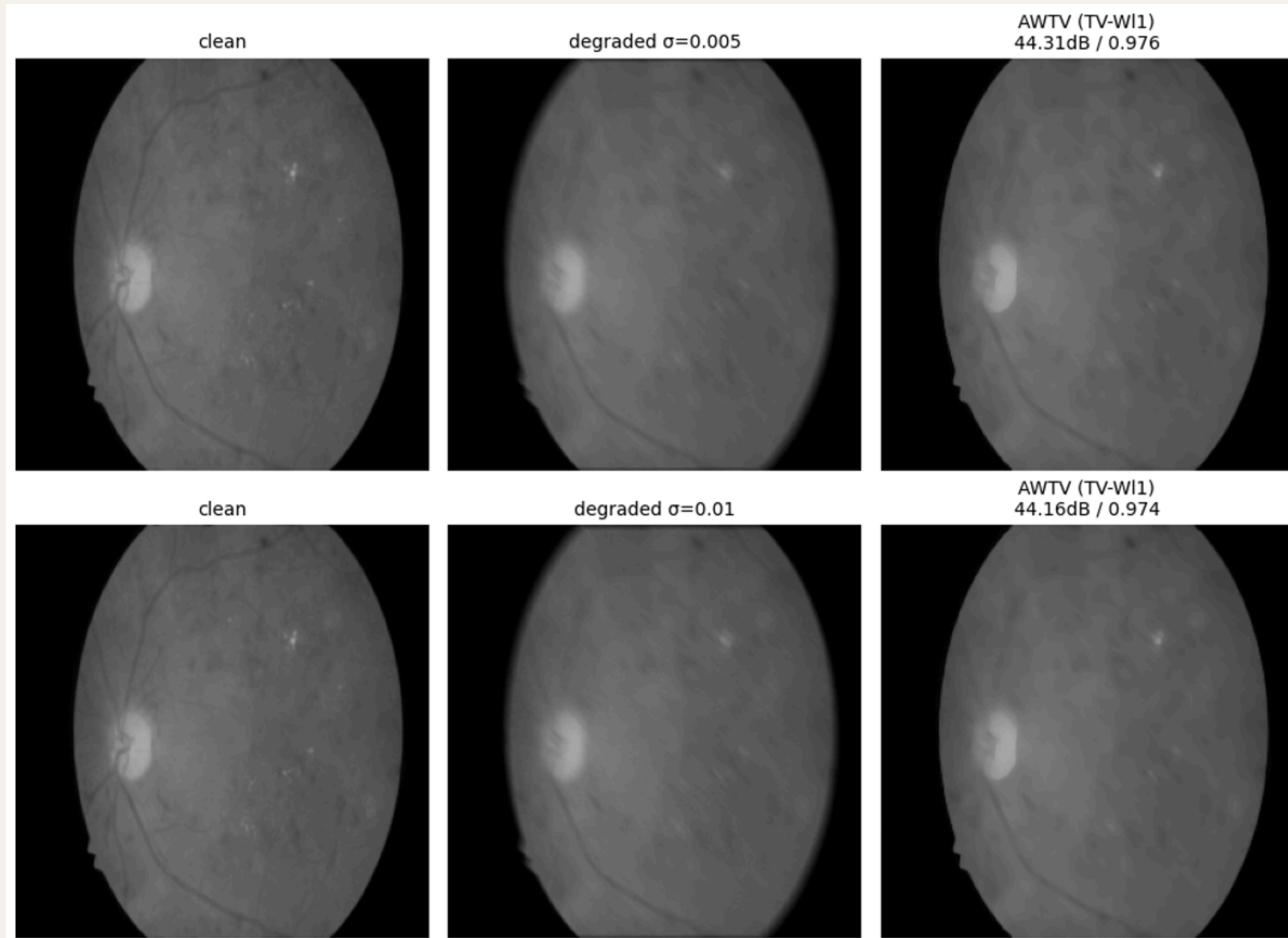
- Performance metrics drop while standard deviation increases (SSIM std rises from 0.013 to 0.091).
- Key Takeaway: Image quality degrades and becomes significantly less consistent across the dataset at higher noise levels.

λ Optimization Dynamics

- Chosen λ increases with noise (0.001 $\rightarrow$ 0.02), aligned with theoretical expectations.
- Grid Limitation: At extreme noise levels, the selected λ hits the grid boundaries (minimum at low noise, maximum at high noise).
- Future Work: Expanding the search grid could further refine the optimum, though it does not affect the validity of the observed trend.

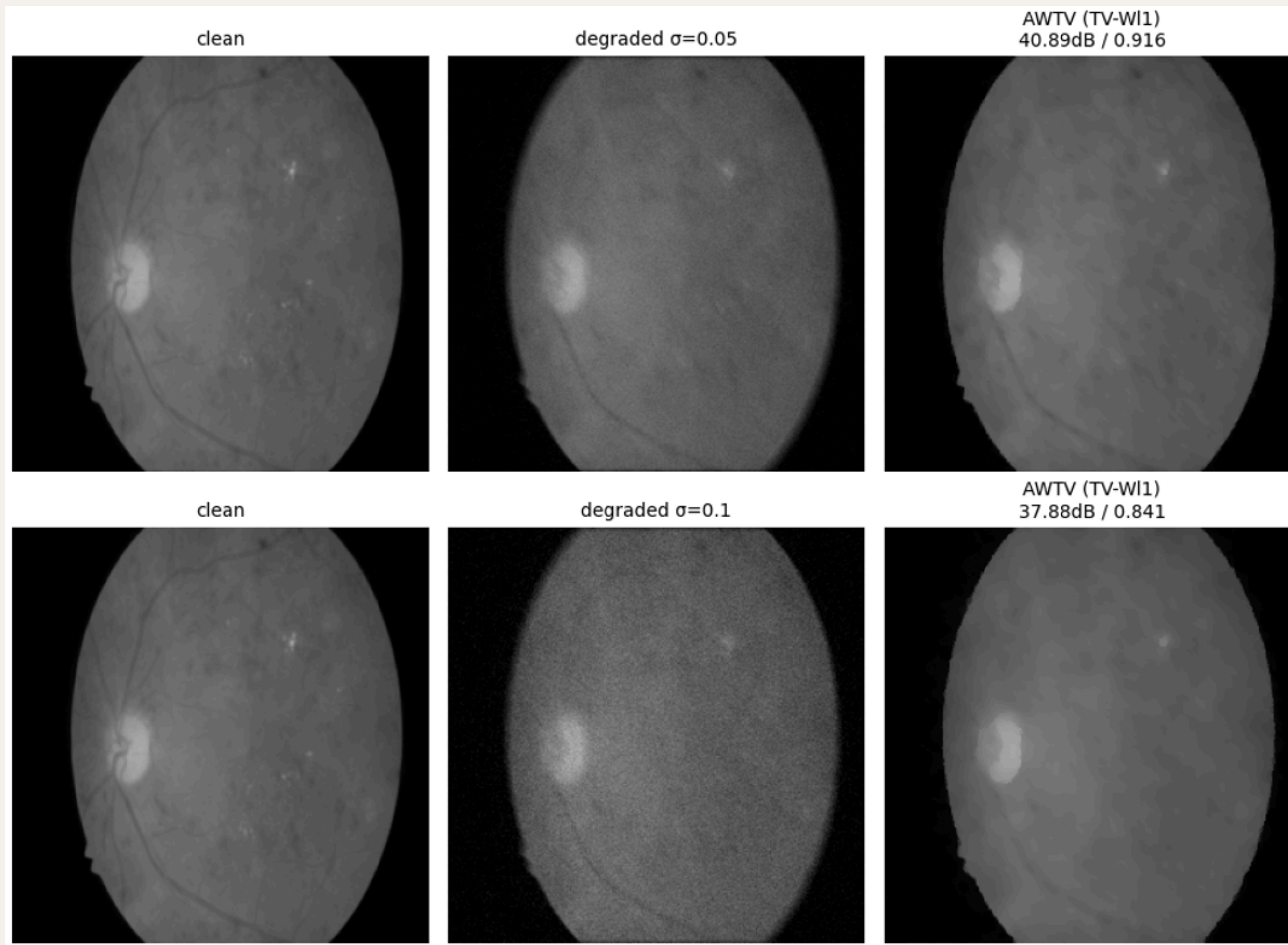
Adaptive Weighted TV- Visual results

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Adaptive Weighted TV- Visual results

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Edge-Preserving Regularization Performance

- Expected Trend: Image quality decreases and variance increases as noise levels rise.
- Mechanism: Adaptive edge-protecting weights (computed from an intermediate image and held fixed) enable strong regularization in flat regions while preserving crucial vessel structures.
- λ Dynamics: Selected λ scales with noise (0.005 --> 0.05). Internal grid values are optimal for low/medium noise; only the highest noise level reaches the grid boundary.

Comparative Analysis: Weighted TV vs. FISTA-Wavelet

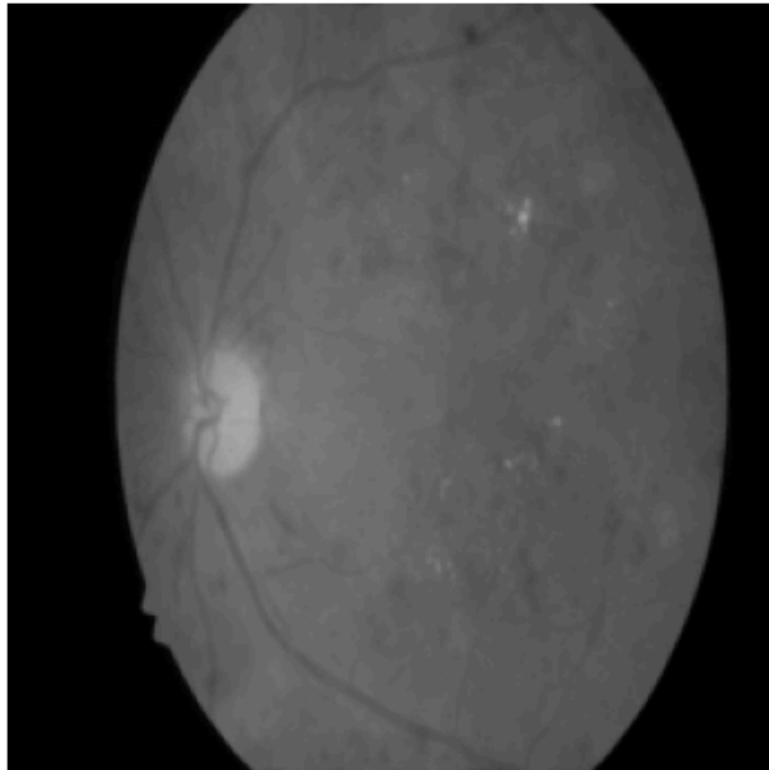
- PSNR Advantage: Adaptive Weighted TV consistently outperforms FISTA-Wavelet across all noise levels.
- Widening Margin: The performance gap expands significantly at medium/high noise, reaching up to +4.7 dB at $\sigma = 0.05$.
- SSIM Improvement: The SSIM advantage grows alongside noise levels, peaking at +0.05 at $\sigma = 0.1$.
- Key Takeaway: Under high noise conditions, an edge-aware prior preserves fine vessel structures far better than global wavelet shrinkage.

Noise	λ	PSNR	SSIM
0.005	0.005	42.07 ± 3.72	0.964 ± 0.022
0.01	0.005	41.93 ± 3.67	0.962 ± 0.022
0.05	0.01	39.27 ± 3.32	0.917 ± 0.044
0.1	0.05	36.69 ± 3.01	0.848 ± 0.089

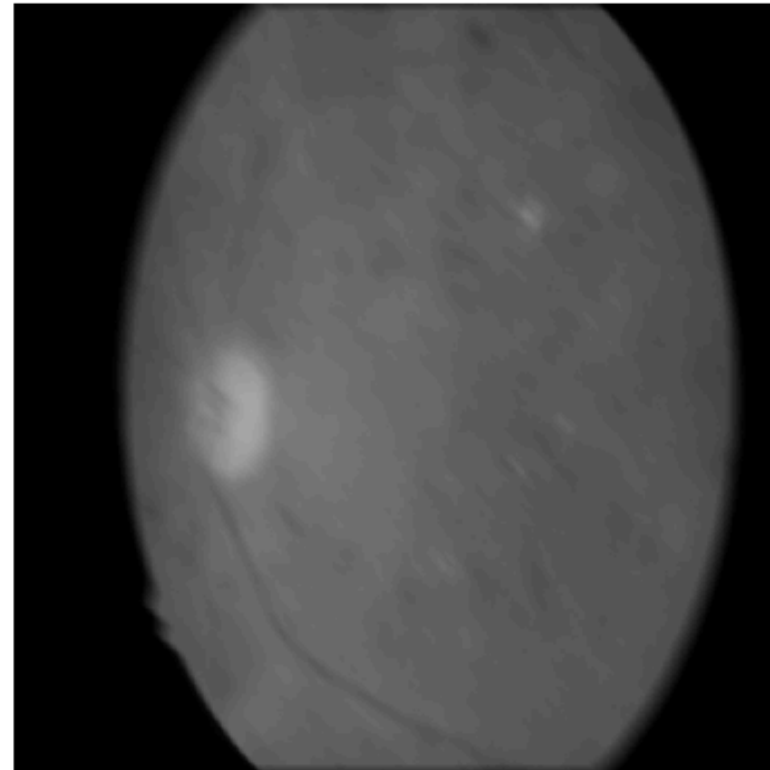
The weighted TV is better at every noise level. And the gap grows with the noise

UNet - Visual results

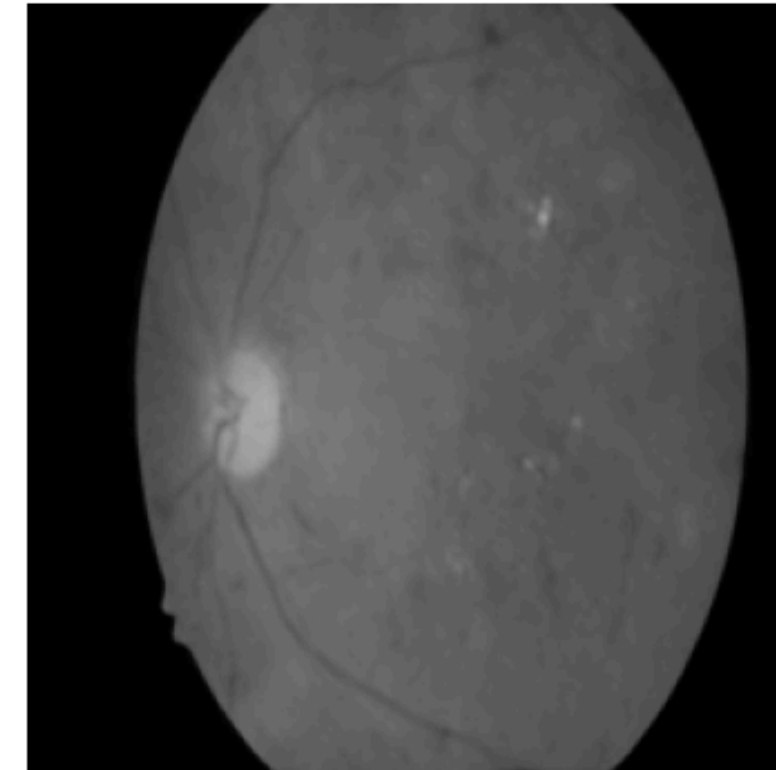
Ground truth



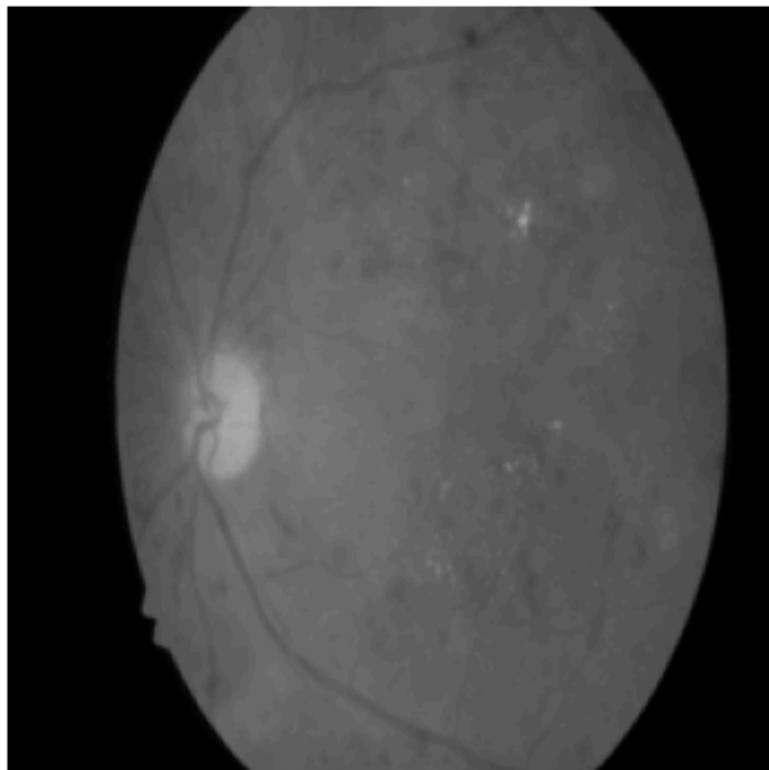
Degraded input
noise=0.005



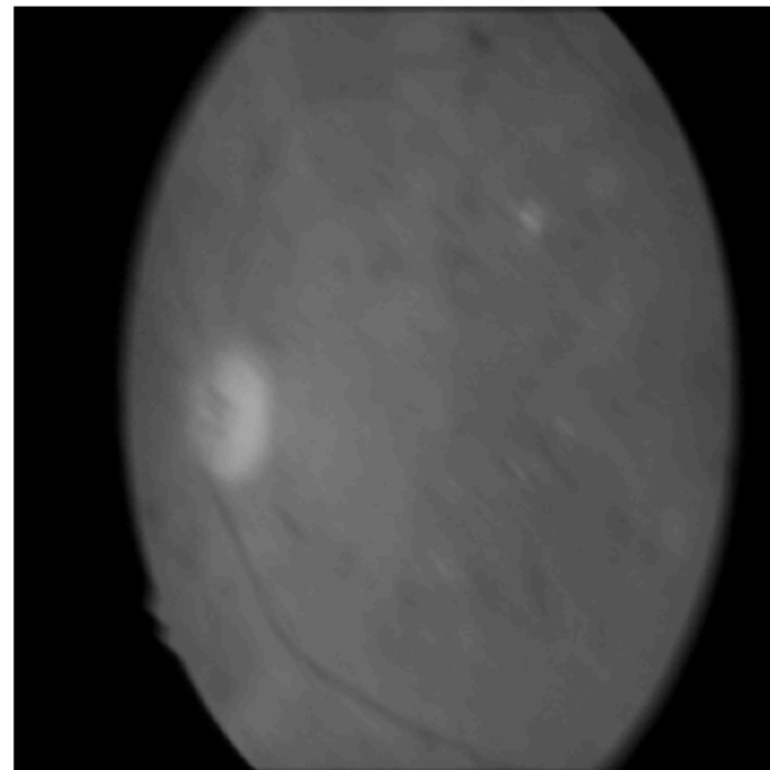
UNet reconstruction



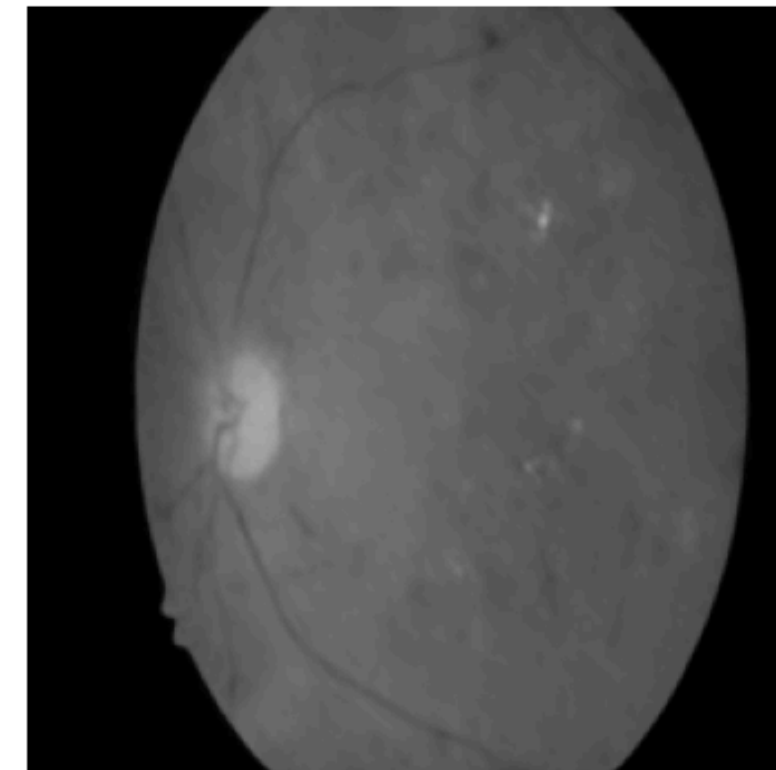
Ground truth



Degraded input
noise=0.01

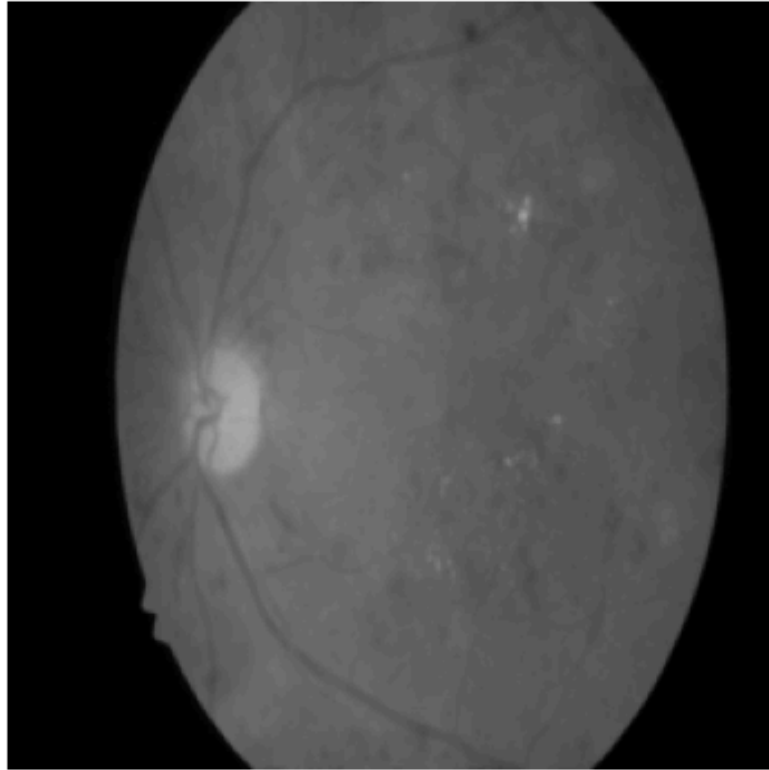


UNet reconstruction

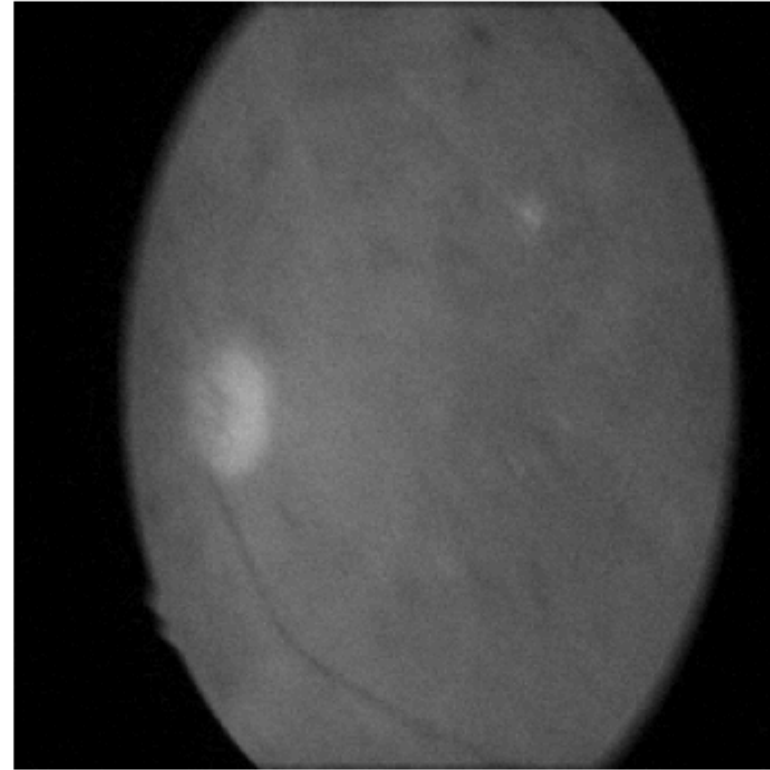


UNet - Visual results

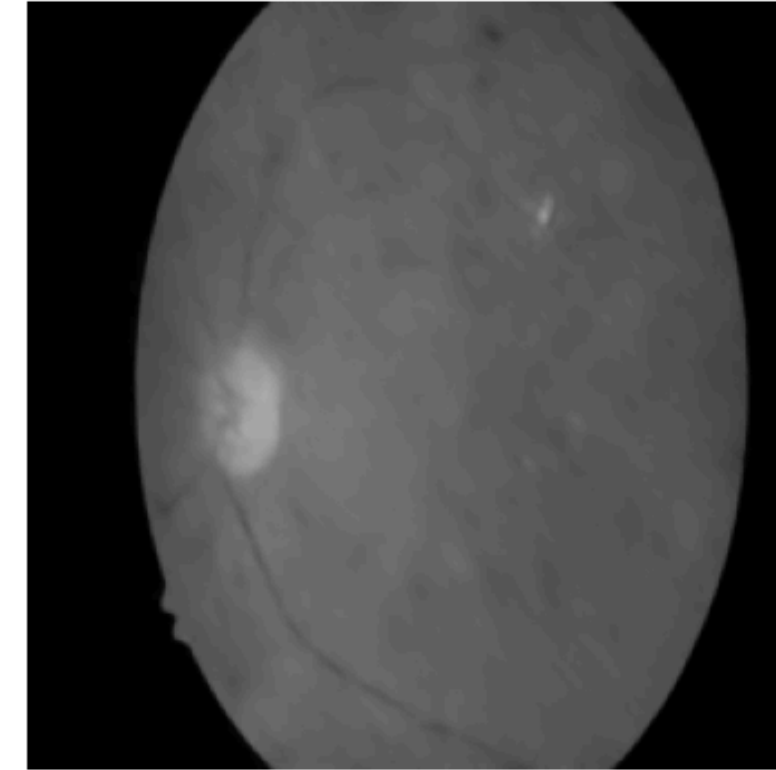
Ground truth



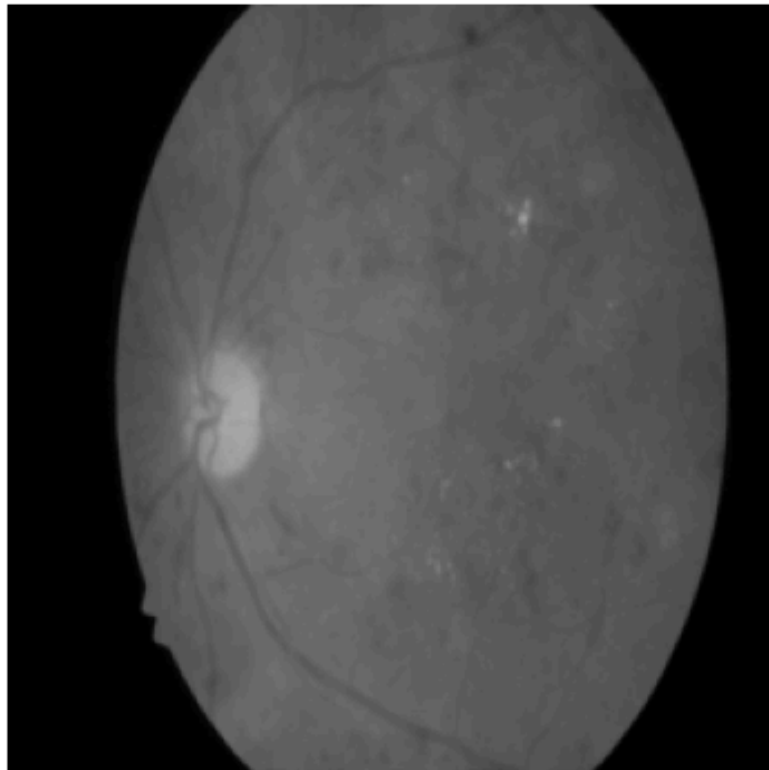
Degraded input
noise=0.05



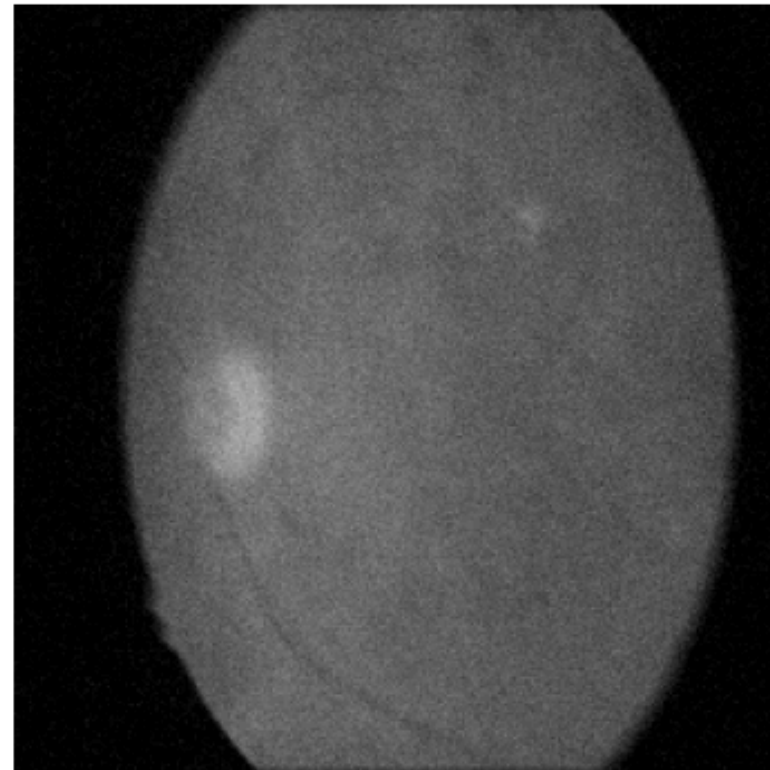
UNet reconstruction



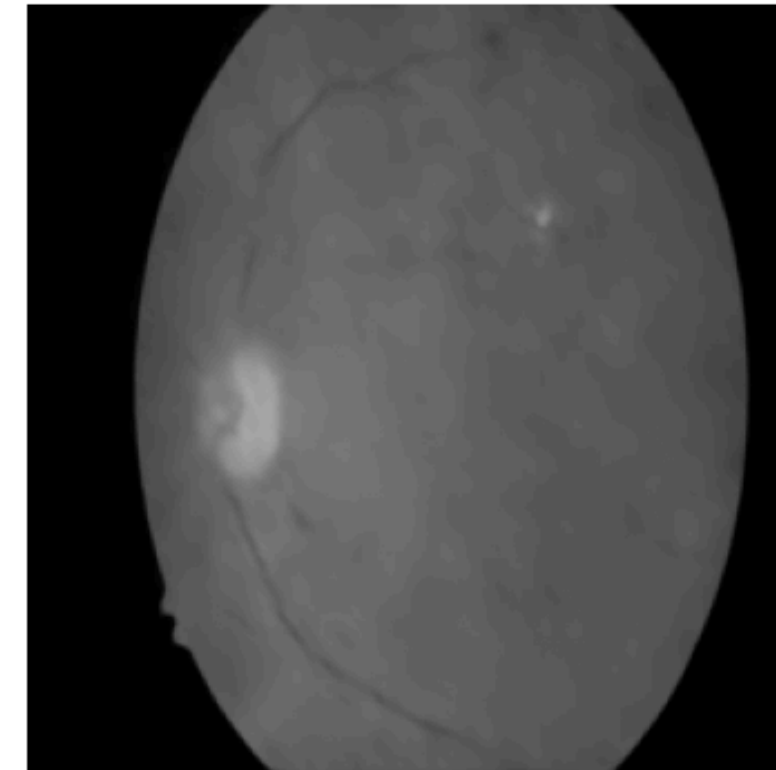
Ground truth



Degraded input
noise=0.1



UNet reconstruction

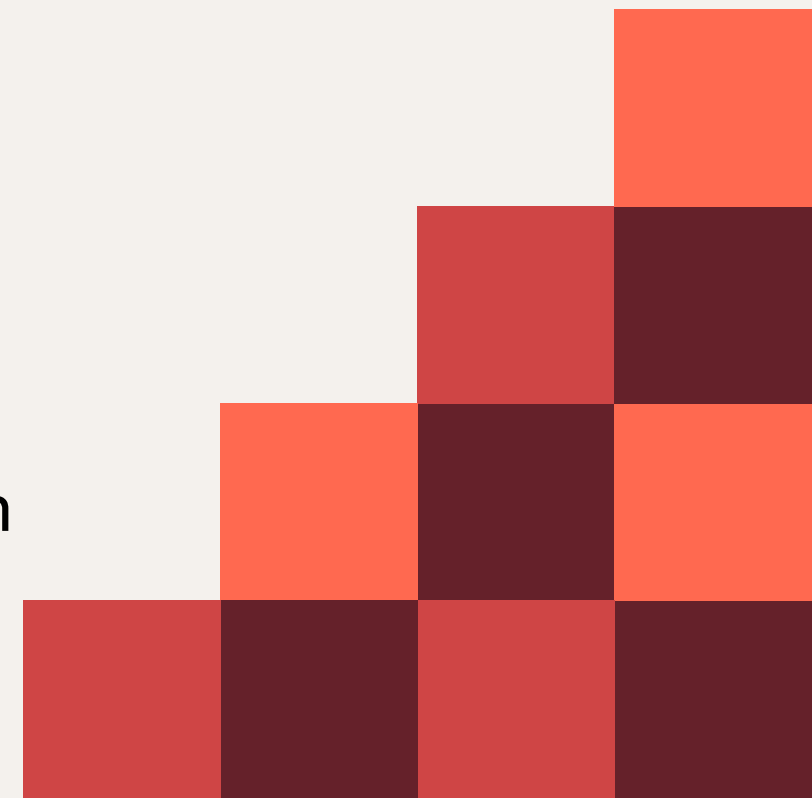


UNet - Numerical results

Noise	PSNR	SSIM
0.005	46.34	0.98
0.01	45.43	0.98
0.05	42.16	0.96
0.1	40.61	0.95

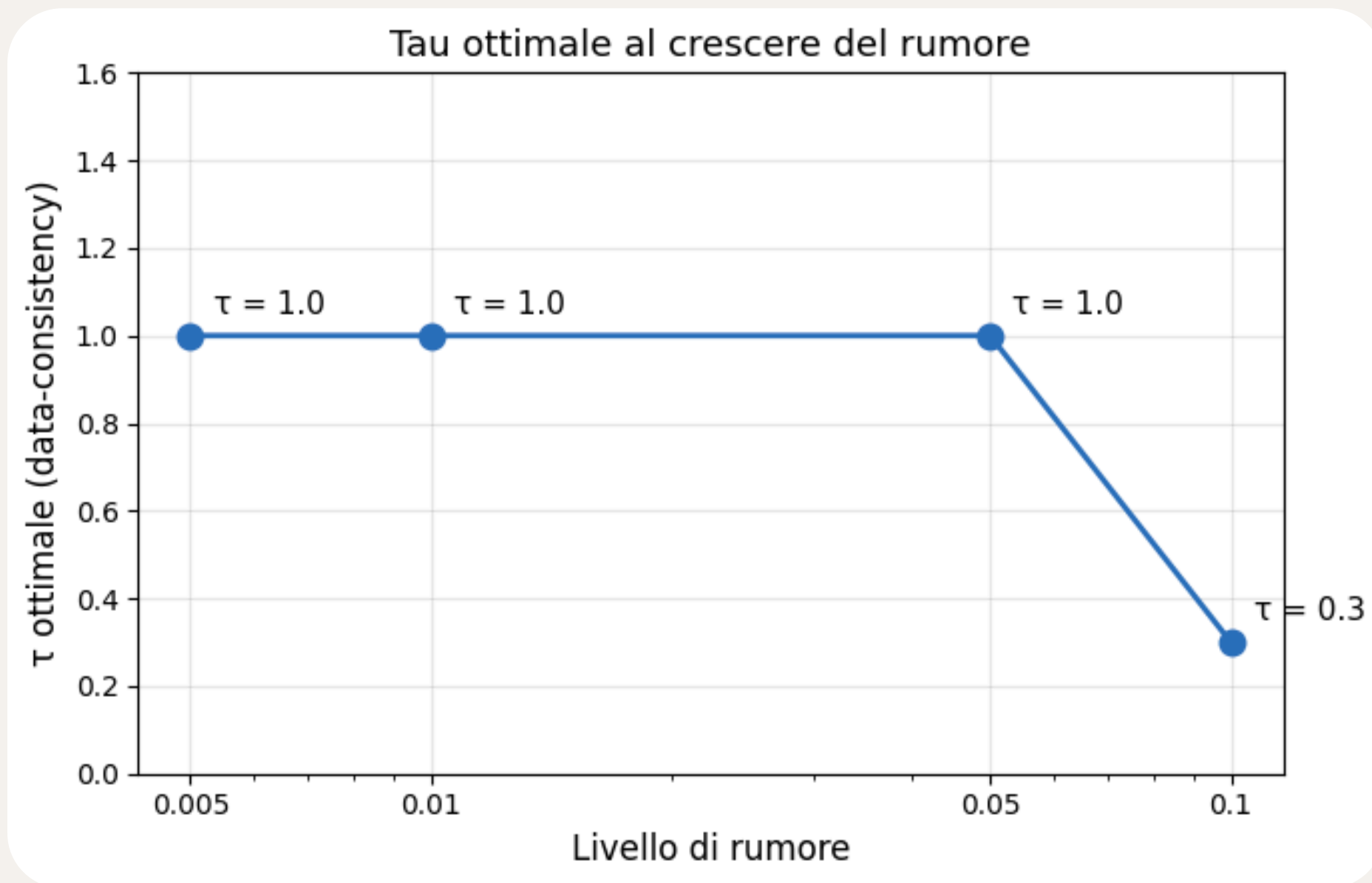
DiffPIR - Experimental Setup

- Evaluation on the test dataset (256x256, grayscale)
- **Degradation:** motion blur (45° angle, kernel size 9) + Gaussian noise at 4 levels:
 - 0.005
 - 0.01
 - 0.05
 - 0.1
- **Metrics:** PSNR and SSIM, averaged over the 600 images
- The parameter τ was tuned on a separate *validation set*, ensuring an unbiased evaluation



Tuning of the τ parameter

τ controls the balance between *data fidelity* and the *diffusion prior*. It was tuned per noise level on the validation set



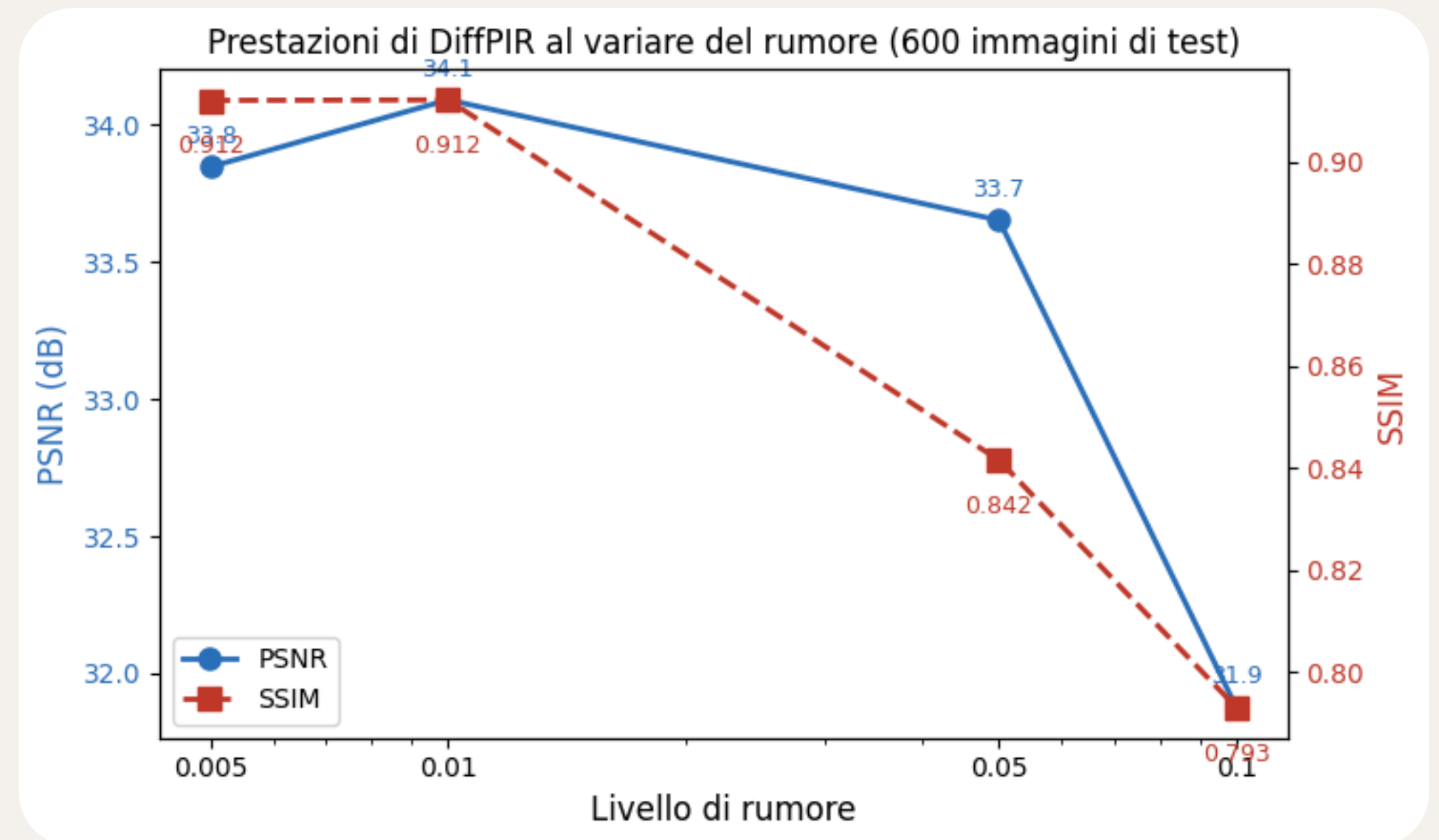
At *low* and *medium* noise the **optimum stays high** ($\tau = 1.0$): the measurement is reliable, so it pays to stay **close to the data**.

At *high* noise the **optimum drops** ($\tau = 0.3$): the measurement is too corrupted, so it is better to **rely on the prior** rather than chase the noise.

DiffPIR - Quantitative results

Performance stays *stable up to medium noise* (PSNR ~34 dB) and *degrades gently at high noise*. The adaptive tuning of τ prevents sharp drops even in the most challenging case.

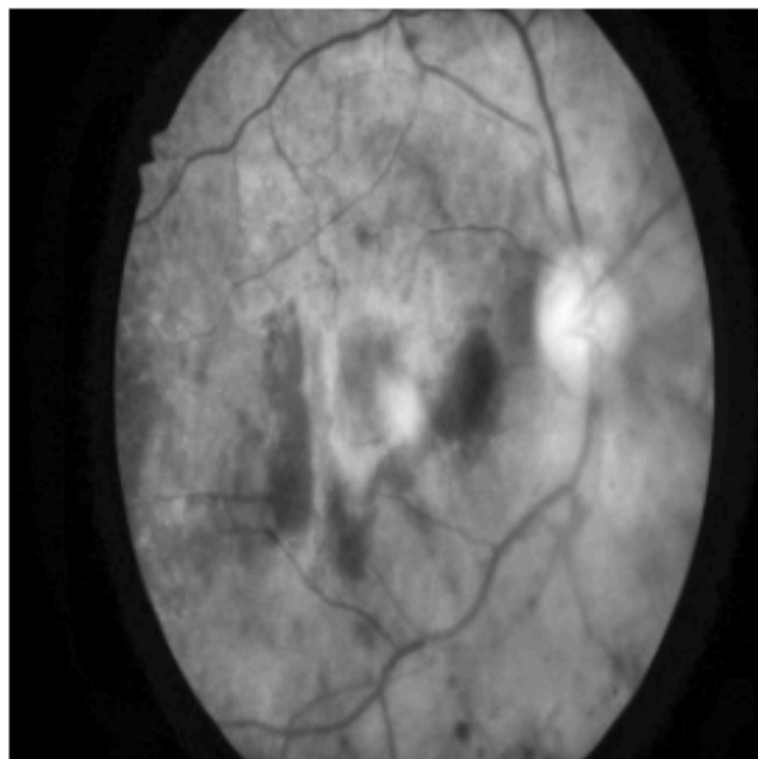
Noise	τ	PSNR	SSIM
0.005	1.0	33.848	0.9121
0.01	1.0	34.092	0.9122
0.05	1.0	33.653	0.8415
0.1	0.3	31.876	0.7930



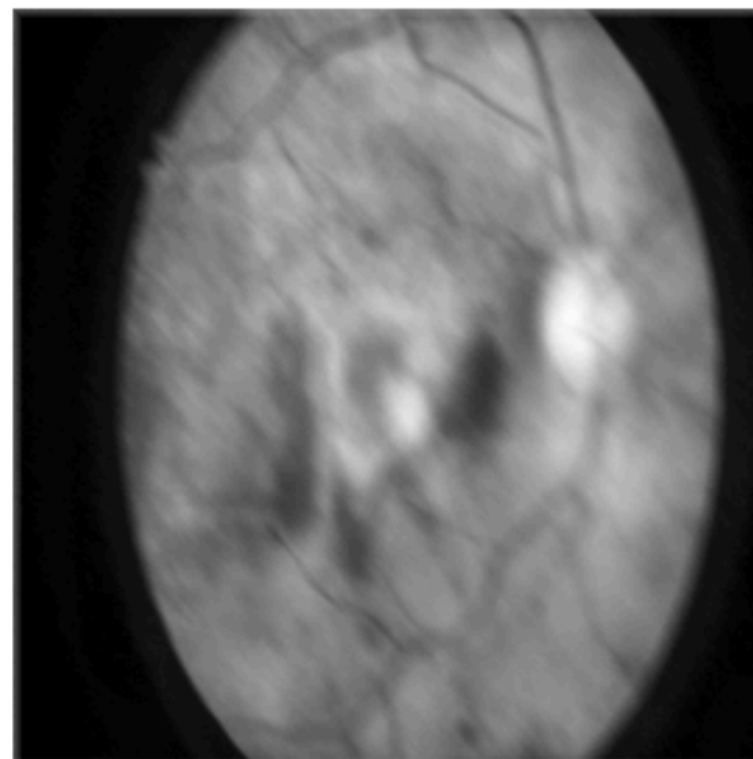
DiffPIR - Visual results

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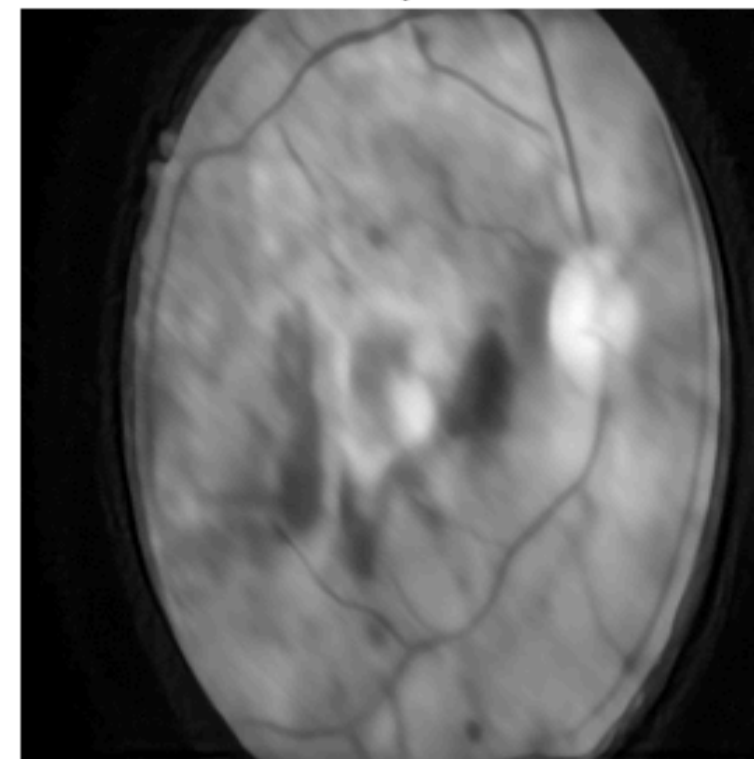
Ground truth



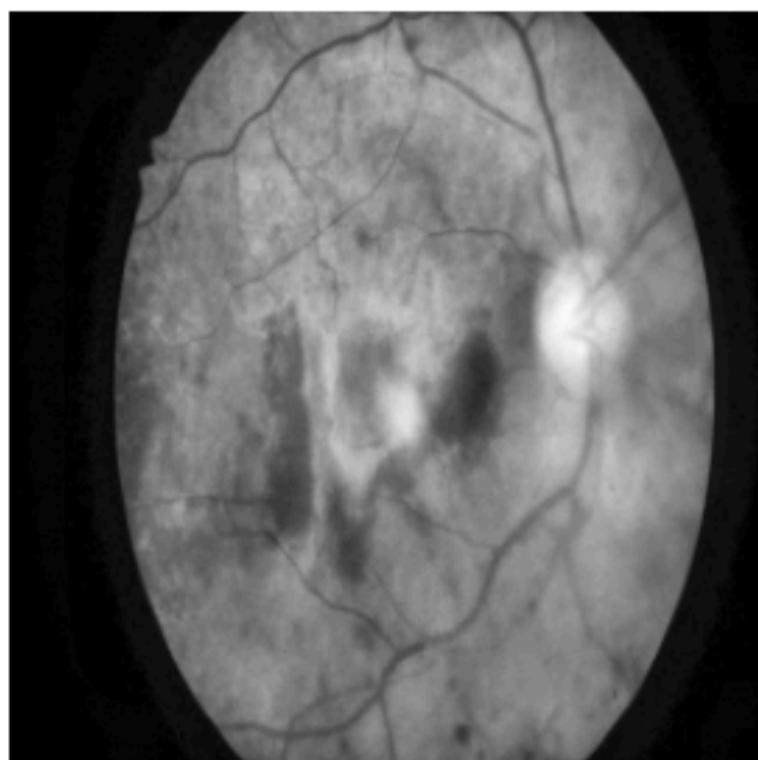
Corrotta
noise=0.005



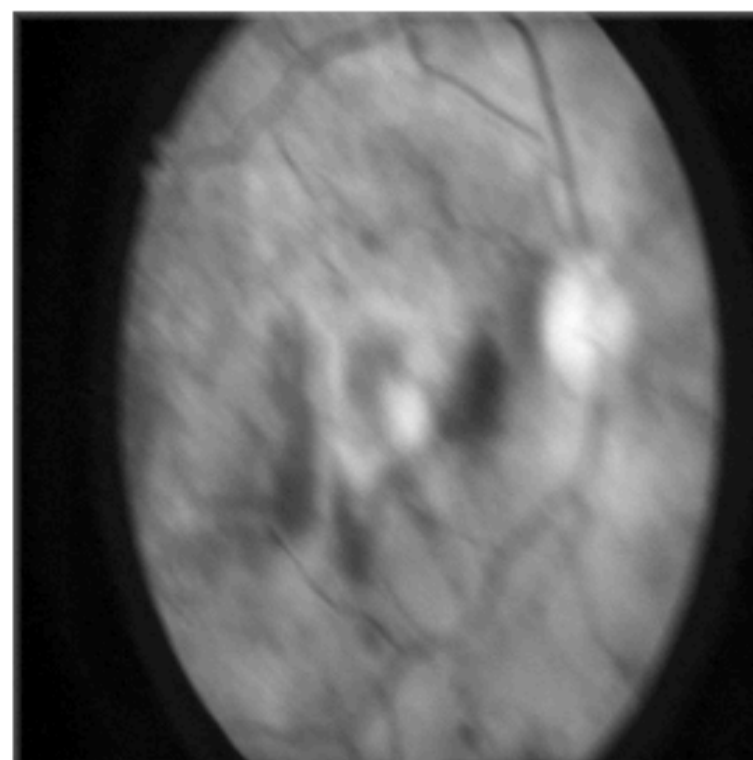
DiffPIR (tau=1.0)
PSNR 32.80 | SSIM 0.879



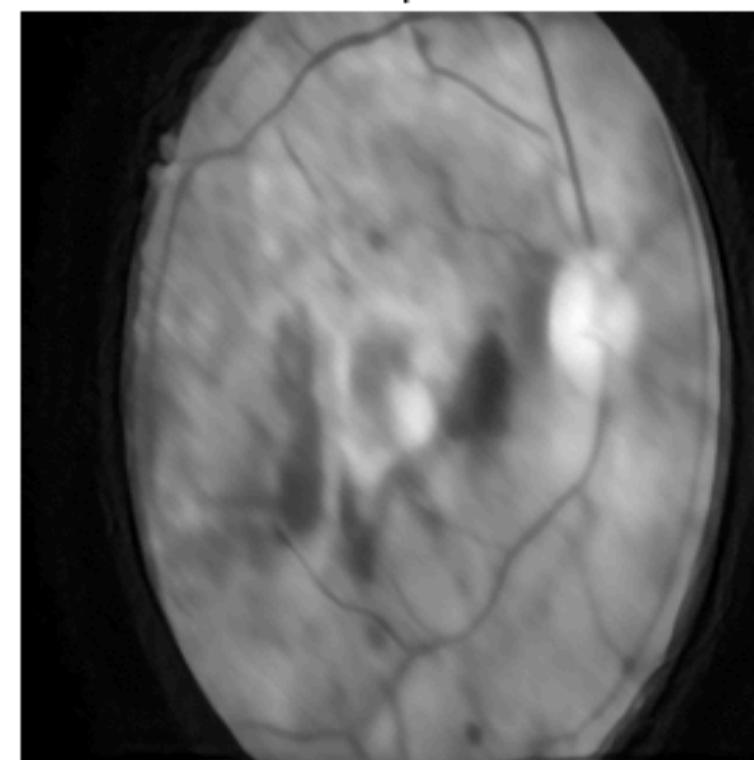
Ground truth



Corrotta
noise=0.01



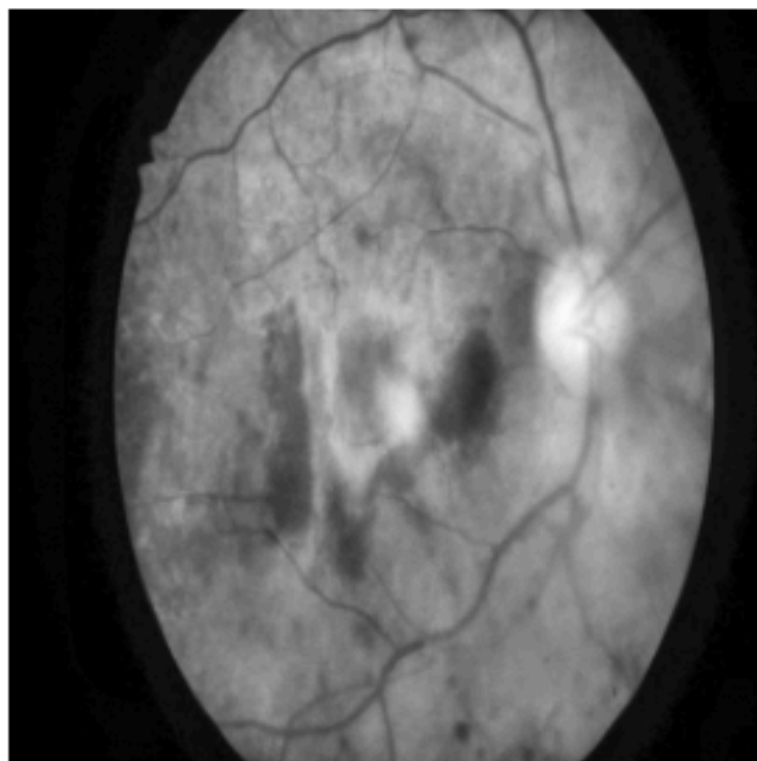
DiffPIR (tau=1.0)
PSNR 33.39 | SSIM 0.885



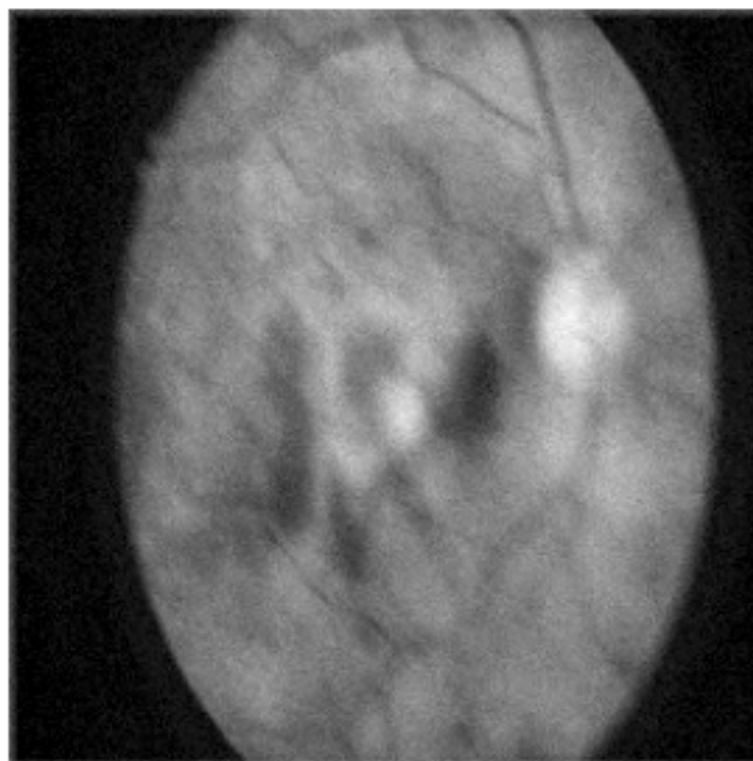
DiffPIR - Visual results

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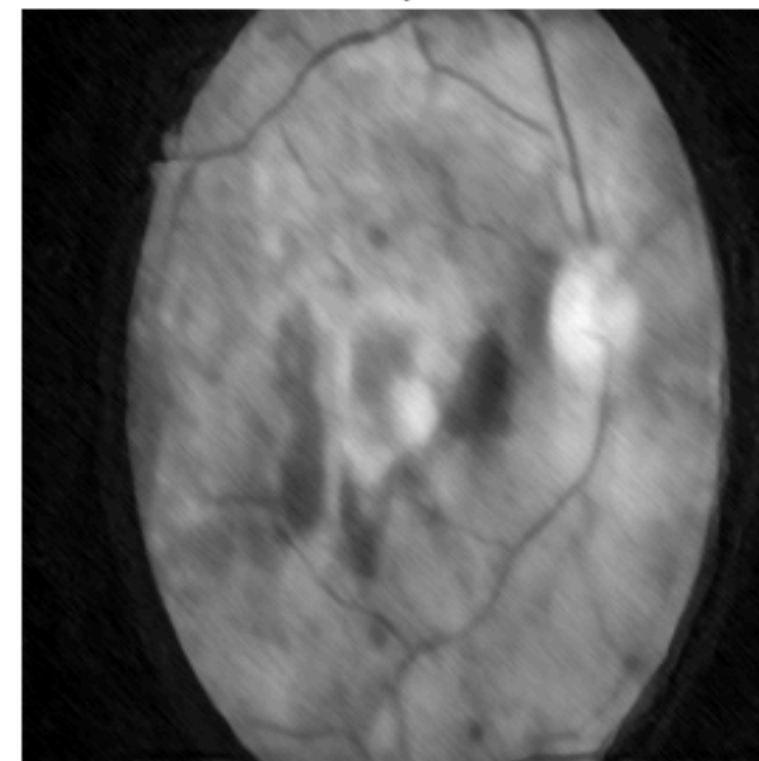
Ground truth



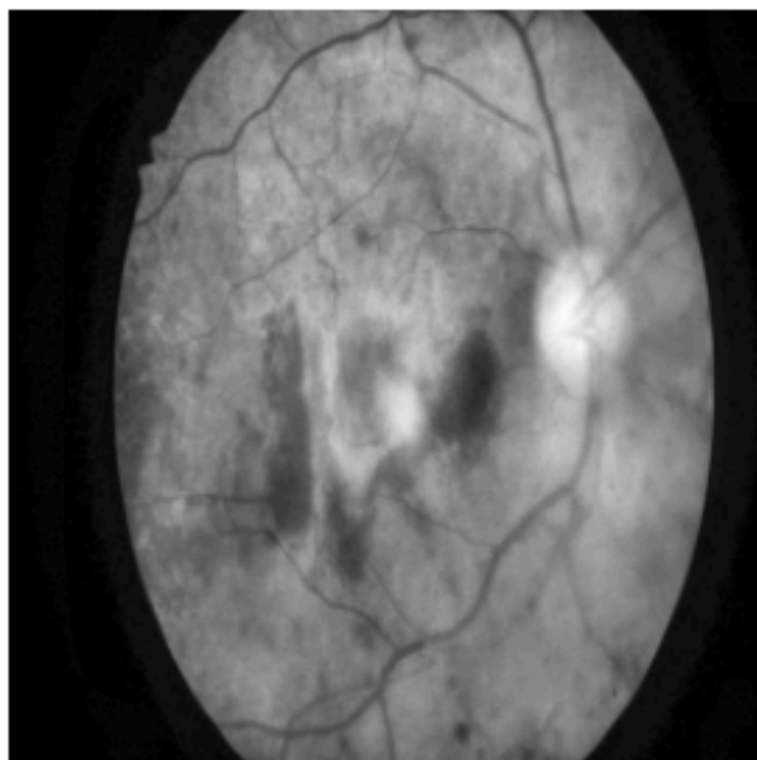
Corrotta
noise=0.05



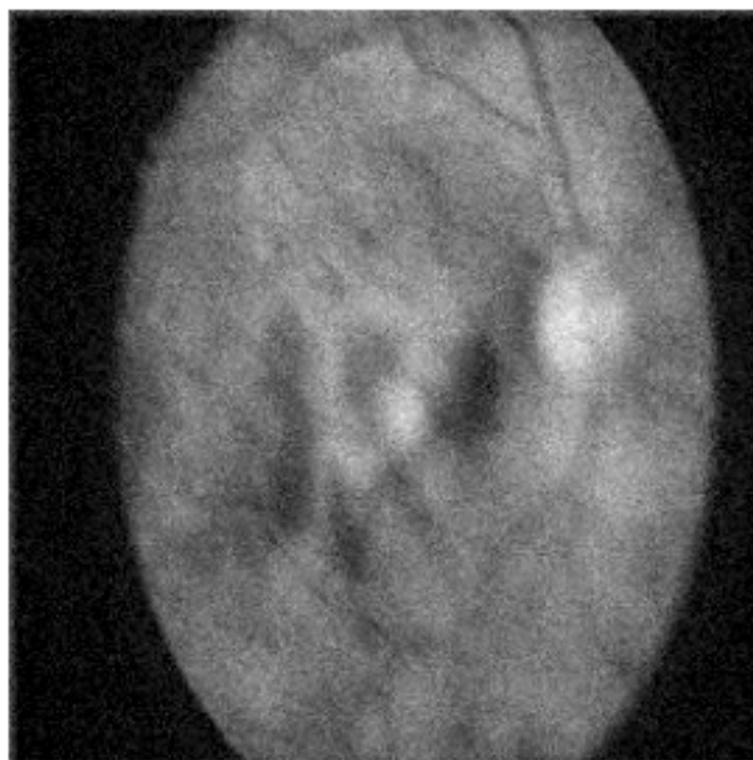
DiffPIR (tau=1.0)
PSNR 33.74 | SSIM 0.851



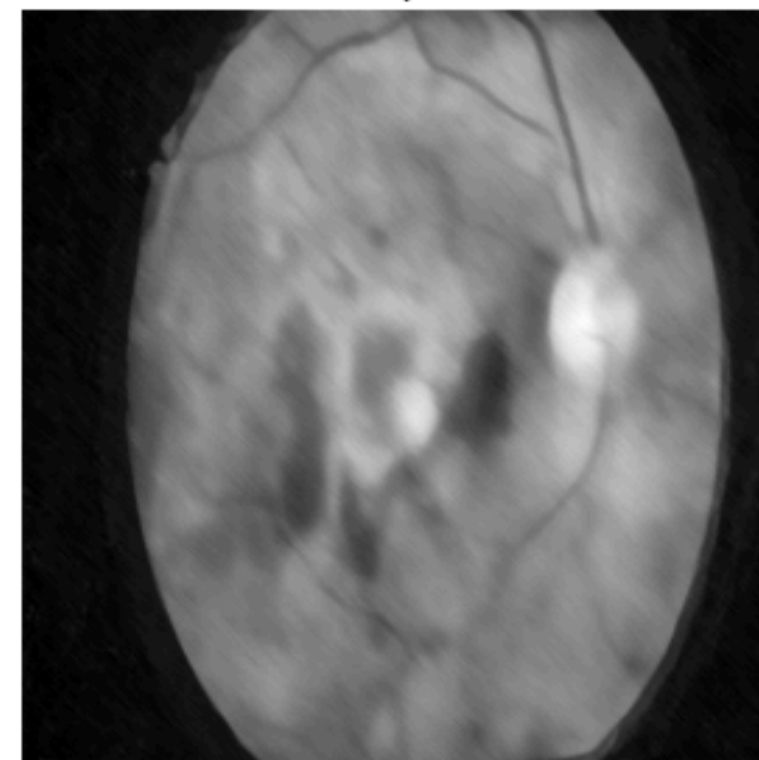
Ground truth



Corrotta
noise=0.1



DiffPIR (tau=0.3)
PSNR 32.50 | SSIM 0.841



DiffPIR - Observations

The reconstructions **recover sharpness** and **vascular structure** even when the measurement is heavily degraded. At high noise the image stays **clean** and **readable**, since the diffusion prior naturally removes the noise

Strengths

- **Robustness to noise**: stable performance across a wide range of levels
- **Plug-and-play approach**: a single prior, reused at every level without retraining
- **Interpretable τ** : it adapts to the noise level

Limitations

- Slight halo at the border, caused by the blur effect in the data-consistency step
- At high noise the finest details (thin vessels) tend to be **smoothed out**
- **Computational cost**: each reconstruction requires several network evaluations

Conclusions

Conclusions

All four methods on the same data: motion blur, 600 test images

Method	$\sigma = 0.005$	$\sigma = 0.01$	$\sigma = 0.05$	$\sigma = 0.1$
FISTA + Wavelet	41.44 / 0.968	40.95 / 0.961	34.53 / 0.891	33.49 / 0.800
AWTV (TV-Wℓ_1)	42.07 / 0.964	41.93 / 0.962	39.27 / 0.917	36.69 / 0.848
UNet (end-to-end)	46.34 / 0.984	45.43 / 0.981	42.16 / 0.964	40.61 / 0.954
DiffPIR (generative)	33.85 / 0.912	34.09 / 0.912	33.65 / 0.842	31.88 / 0.793

PSNR (dB) / SSIM, mean over the 600 test images.

- **FISTA + Wavelet** - simple and convex; strong at low noise, but smooths everything, so it loses thin vessels at high noise.
- **AWTV (TV-W ℓ_1)** - best model-based method; adaptive weights protect the vessel edges and beat FISTA at every level, up to +4.7 dB.
- **UNet** - best PSNR and SSIM overall, but needs training per noise level and does not enforce data consistency.
- **DiffPIR** - a single reusable prior, most robust to noise, but lower PSNR and the heaviest to run.

Bibliography



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